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Waffle slab and structures and method of manufacturing the same

Abstract

A method of manufacturing a waffle slab includes providing a bottom mold separately on columns and exposing joint heads of the columns from the bottom mold; providing first sets of beam bars separately on the bottom mold to be parallel along a first direction; and providing second sets of beam bars separately on the bottom mold to be parallel along a second direction, wherein the second sets b of beam bars are configured to intersect with the first sets of beam bars without interference and together they form accommodation spaces for waffle molds.

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Background/Summary

CROSS-REFERENCE

(1) This application claims priority to Taiwan application No. 111137288 filed on Sep. 30, 2022, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present disclosure relates to a building structure and a method for manufacturing the same, and more particularly to a waffle slab, a structure and a method of manufacturing the same.

BACKGROUND

(3) Waffle slabs are widely used in constructing the floors of high-tech fabs for carrying machines and equipment. As a robust structural system with effective vibration isolation of machines and equipment, waffle slabs are capable of minimizing the impact from various sources such as pumps, fans, piping, ducts within the building, and even earthquakes.

(4) Precast columns are manufactured in a controlled factory setting by pouring concrete into reusable molds, allowing it to solidify and harden, and then transporting the finished columns to the construction site for installation. Precast columns, compared to the columns made on construction sites, have the advantages of environmental stability, no weather considerations, less need for skilled labor, and standardized operation procedures. In addition, on construction sites, mechanical equipment can be used to assemble and lift the precast columns and no external scaffolding is required for workers to assemble said columns.

(5) A construction method of placing waffle slabs on precast columns combines the advantages of both systems. However, since waffle slabs have many perforations and reinforcement structures around the perforations to provide the required strength to the waffle slabs, the above construction method involves settings of the waffle mold, settings of reinforcement ties, stirrups and settings of formwork, which need to be carried out at a certain height above the precast columns. Consequently, such a construction method involves a relatively high level of difficulty and complexity.

(6) Thus, to solve the aforementioned problems, it is desirable to provide a systematic and efficient method of manufacturing a waffle slab on precast columns.

SUMMARY

(7) An embodiment of the present disclosure provides a method of manufacturing a waffle slab on multiple columns at intervals. The method includes providing a bottom mold separately on multiple columns and exposing joint heads of the multiple columns from a top surface of the bottom mold; providing multiple first sets of beam bars separately on the bottom mold so that the multiple first sets of beam bars are parallel to each other along a first direction, wherein the multiple first sets of beam bars each includes: multiple first rebar cages disposed along the first direction at intervals, and multiple first upper rebars and multiple first lower rebars that both penetrate through the inside of the multiple first rebar cages and are respectively affixed to the upper portions and lower portions of the multiple first rebar cages; providing multiple second sets of beam bars separately on the bottom mold so that the multiple second sets of beam bars are parallel to each other along a second direction, the second direction being different from the first direction, wherein the multiple second sets of beam bars each includes: multiple second rebar cages separately disposed along the second direction; and multiple second upper rebars penetrating through the inside of the multiple second rebar cages and affixed to the upper portions of the multiple the second rebar cages, wherein the multiple second upper rebars are attached to parts of the multiple first upper rebars that are located between two of the multiple first rebar cages; and providing and penetrating multiple second lower rebars through an inside of the multiple second rebar cages and affixing the multiple second lower rebars to the lower portions of the multiple the second rebar cages; wherein the multiple second sets of beam bars are configured to intersect with the multiple first sets of beam bars without interference, so that the multiple first sets of beam bars and the multiple second sets of beam bars together form multiple accommodation spaces therebetween.

(8) Another embodiment of the present disclosure provides a waffle slab structure, including: a bottom mold disposed on multiple columns disposed at intervals, wherein a joint head of each of the multiple columns is exposed from a top surface of the bottom mold; multiple first sets of beam bars disposed separately on the bottom mold that are parallel to each other along a first direction, wherein the multiple first sets of beam bars each includes: multiple first rebar cages disposed along the first direction at intervals, and multiple first upper rebars and multiple first lower rebars that both penetrate through an inside of the multiple first rebar cages and are respectively affixed to the upper portions and lower portions of the multiple first rebar cages; multiple second sets of beam bars disposed separately on the bottom mold that are parallel to each other along a second direction, the second direction being different from the first direction, wherein the multiple second sets of beam bars each includes: multiple second rebar cages separately disposed along the second direction; and multiple second upper rebars and multiple second lower rebars that both penetrate through an inside of the multiple second rebar cages and are respectively affixed to the upper portions and lower portions of the multiple second rebar cages, wherein the multiple second upper rebars are attached to parts of the multiple first upper rebars that are located between two of the multiple first rebar cages; wherein the multiple second sets of beam bars are configured to intersect with the multiple first sets of beam bars without interference, so that the multiple first sets of beam bars and the multiple second sets of beam bars together form multiple accommodation spaces therebetween; and multiple waffle molds are disposed in the multiple accommodation spaces.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The drawings described below are for illustrative purposes only, and are not intended to limit the scope of the present disclosure in any way.

(2) FIG. **1** is a schematic view showing an embodiment of the present disclosure.

(3) FIG. **2** is a schematic view of the rail shown in FIG. **1**;

(4) FIG. **3** is a schematically perspective view of a first set of beam bars, in accordance with the embodiment of the present disclosure;

(5) FIG. **4** is a schematic view showing a lifter hoisting multiple sets of first beam bars, in accordance with the embodiment of the present disclosure;

(6) FIG. **5** is a further schematic view showing the embodiment of the present disclosure;

(7) FIG. **6** is a schematically perspective view of a second set of beam bars, in accordance with the embodiment of the present disclosure;

(8) FIG. **7** is a still further schematic view showing the embodiment of the present disclosure;

(9) FIG. **8** is a partially enlarged schematic view of the structure shown in FIG. **7**;

(10) FIG. **9** is a still further schematic view showing the embodiment of the present disclosure;

(11) FIG. **10** is a still further schematic view showing the embodiment of the present disclosure;

(12) FIG. **11** is a still further schematic view showing the embodiment of the present disclosure;

(13) FIG. **12** is a schematic ally perspective view of a column tie bar, in accordance with the embodiment of the present disclosure;

(14) FIG. **13** is an enlarged schematic view showing a column joint, in accordance with the embodiment of the present disclosure;

(15) FIG. **14** is a still further schematic view showing the embodiment of the present disclosure;

(16) FIG. **15** is a schematic view of a waffle mold, in accordance with the embodiment of the present disclosure;

(17) FIG. **16** is a still further schematic view showing the embodiment of the present disclosure;

(18) FIG. **17** is a still further schematic view showing the embodiment of the present disclosure;

(19) FIG. **18** is a schematically perspective view showing the embodiment of the present

disclosure;
(20) FIG. **19** is a schematic bottom view of the structure shown in FIG. **18**; and
(21) FIG. **20** is a partial cross-sectional view of FIG. **18** along line A-A.

DETAILED DESCRIPTION OF THE DISCLOSURE

(22) Embodiments of the present disclosure, including various examples, will be described in detail below with reference to the accompanying drawings. It should be noted that the content of the implementation mode of this case is only used to illustrate a specific aspect of this disclosure, and is not intended to limit the scope of disclosure requested in this case.
(23) FIG. **1** is a schematic view showing an embodiment of the present disclosure, which shows that a bottom mold **1** is provided and disposed on multiple columns **10** arranged at intervals. As shown in FIG. **1**, nine columns **10** are arranged at intervals and the tops of the nine columns **10** define a rectangular plane. The bottom mold **1** is placed on top of these columns **10**, and is coplanar with the rectangular plane. The joint head **12** of each of the columns **10** is exposed from a top surface of the bottom mold **1**. In this embodiment, the plane has a first direction D**1** and a second direction D**2** in which the first direction D**1** and the second direction D**2** are substantially perpendicular to each other. In some embodiments, the columns **10** are precast columns. As shown in FIG. **1**, multiple rails **2** are arranged on the bottom mold **1** at intervals along the second direction D**2**.
(24) FIG. **2** is a schematic view of the suspension rail **2** shown in FIG. **1**. Each of the rails **2** includes an elongated trough-shaped body **20** and multiple supports **22** disposed on the elongated trough-shaped body **20** at intervals. The edge of the elongated trough-shaped body **20** includes two opposite bent portions **23**. An opening **24** is formed between the bent portions **23**, and the multiple supports **22** are located on the opposite side of the opening **24** with respect to the bent portions **23**. In addition, the openings **24** of the elongated trough-shaped bodies **20** of the multiple rails **2** are configured to face the bottom mold **1** (see FIG. **1**).
(25) FIG. **3** is a schematically perspective view of a first set of beam bars **3**, in accordance with the embodiment of the present disclosure. The first sets of beam bars **3** includes multiple first rebar cages **30**, multiple first upper rebars **32** and multiple first lower rebars **34**. The multiple first upper rebars **32** and the multiple first lower rebars **34** penetrate through an inside of the multiple first rebar cages **30** and are affixed to the upper and lower portions of the multiple first rebar cages **30**, respectively. The multiple first rebar cages **30** are spaced apart from each other and form multiple gaps W**1**. As shown in FIG. **3**, three first upper rebars **32** are fixed to the upper portions of the multiple first rebar cages **30** from the inside of the multiple first rebar cages **30**, and two first lower rebars **34** are fixed to the lower portions of the multiple first rebar cages **30** from the inside of the multiple first rebar cages **30**.
(26) In this embodiment, each of the first set of beam bars **3** includes multiple first waist bars **36** and multiple first tie bars **38**. The multiple first waist bars **36** are located between the multiple first upper rebars **32** and the multiple first lower rebars **34**, and are arranged to be substantially parallel to the multiple first upper rebars **32** and the multiple first lower rebars **34**. The multiple first tie bars **38** are used to fix the multiple first waist bars **36** for enhancing the strength of the first set of beam bars **3**. As shown in FIG. **3**, there are four first waist bars **36** in this embodiment, two of which are close to the multiple first upper rebars **32**, and are fixed by the upper first waist bars **36**, the other two of which are close to the multiple first lower rebars **34** and are fixed by the lower first waist bars **36**.
(27) FIG. **4** is a schematic view showing a lifter **5** hoisting multiple sets of first sets of beam bars **3**, in accordance with the embodiment of the present disclosure. FIG. **4** shows that multiple first sets of beam bars **3** arranged at intervals are lifted simultaneously by a lifter **5**, and the lifter **5** is configured so that the heights of the multiple first sets of beam bars **3** on the lifter **5** are different. In this embodiment, the five first sets of beam bars **3** are hoisted by hanging ropes **50** of the lifter **5**, and the heights of the five first sets of beam bars **3** gradually increase from the left side to the right

side of FIG. **4**. By this disposition of the first sets of beam bars **3** of different heights as shown in FIG. **4**, more beam bars **3** can be hung on the lifter **5**. In prior art, the first sets of beam bars **3** are arranged at the same height on the lifter **5** resulting in the lifter **5** needing to be raised and lowered repeatedly for placing the first sets of beam bars **3**. In contrast, in this embodiment, with a slow descent in one action, the first sets of beam bars **3** are placed on the bottom mold **1** in sequence and thus the efficiency is improved. In some embodiments, the suspension rail **2** shown in FIG. **1** and FIG. **2** may also be transported to the bottom mold **1** by using the lifter **5**.
(28) FIG. **5** is a further schematic view showing the embodiment of the present disclosure. FIG. **5** shows that multiple first sets of beam bars **3** are sequentially arranged parallel to each other along the first direction D**1** on the bottom mold **1** and arranged at intervals at a predetermined distance.
(29) FIG. **6** is a schematically perspective view of a second set of beam bars **4**, in accordance with the embodiment of the present disclosure. The second set of beam bars **4** includes multiple second rebar cages **40**, **40***a* arranged at intervals and multiple second upper rebars **42**. In this embodiment, there are three second upper rebars **42** sequentially penetrating and supporting multiple second rebar cages **40**, **40***a* from below. The multiple second rebar cages **40**, **40***a* are separated from each other by a gap W**2**. In addition, multiple stirrups in the second reinforcement cage **40***a* at a far end (for example, the left side of FIG. **6**) is denser than the other second reinforcement cages **40**. At the left ends of second upper rebars **42**, multiple connectors **43** are provided for connecting the corresponding second set of beam bars **4**.
(30) FIG. **7** is a still further schematic view showing the embodiment of the present disclosure. FIG. **7** shows that multiple second sets of beam bars **4** are arranged along a second direction D**2** on the bottom mold **1** and are parallel with each other and at intervals of a predetermined distance. Each of the multiple second sets of beam bars **4** are formed of two sections in serial connected by the multiple connectors **43** shown in FIG. **6**.
(31) As shown in FIG. **7**, multiple second sets of beam bars **4** are configured to intersect with multiple first sets of beam bars **3**, and do not interfere with each other. They are intersected at the gaps W**1** between the multiple first rebar cages **30** and at the gaps W**2** between the multiple second rebar cages **40**. The multiple first sets of beam bars **3** and the multiple second sets of beam bars **4** form an interlaced structure and form multiple accommodating spaces S therein. In the embodiment shown in FIG. **1** and FIG. **2**, multiple second sets of beam bars **4** are provided on the multiple supports **22** of the multiple rails **2** (see FIG. **2**), which are on the bottom mold **1**.
(32) FIG. **8** is a partially enlarged schematic view of the structure shown in FIG. **7**. FIG. **8** shows multiple second upper rebars **42** that sequentially penetrate multiple second rebar cages **40**, and overlap multiple first upper rebars **32** of the first sets of beam bars **3**. The overlapped portions of the multiple first upper rebars **32** are between two adjacent first rebar cages **30** of the first sets of beam bars **3**. FIG. **8** shows that multiple connectors **43** are provided to connect the multiple second upper rebars **42** in a section of the second set of beam bars **4** to those of the other section of the second set of beam bars **4**. In this embodiment, the pre-folded second rebar cage **40***a* in the section of the second set of beam bars **4** is stretched onto the other section of the multiple second upper rebars **42***a* of the second set of beam bars **4** on the left side of FIG. **8**.
(33) FIG. **9** is a still further schematic view showing the embodiment of the present disclosure. FIG. **9** shows multiple second lower rebars **44** penetrating through the multiple second rebar cages **40**, **40***a* and fixed to the lower portions of the multiple second rebar cages **40**, **40***a*.
(34) FIG. **10** is a still further schematic view showing the embodiment of the present disclosure. FIG. **10** shows multiple second waist bars **46** affixed to the multiple second rebar cages **40**, **40***a* of the second set of beam bars **4** and located between the multiple second upper rebars **42** and the multiple second lower rebars **44**. The multiple second waist bars **46** are approximately parallel to the multiple second upper rebars **42** and the multiple second lower rebars **44**.
(35) FIG. **11** is a still further schematic view showing the embodiment of the present disclosure. FIG. **11** shows multiple second tie bars **48** used to fix a pair of second waist bars **46** respectively

located on opposing sides of the second set of beam bars **4**. In some embodiments, the multiple second tie bars **48** are U-shaped to hook the pairs of second waist bars **46** via the bent ends of the multiple second tie bars **48**.

(36) FIG. **12** is a schematically perspective view of a column tie bar **6**, in accordance with the embodiment of the present disclosure. The column tie bar **6** includes multiple first rectangular structures **60** arranged parallel to each other and multiple second rectangular structures **62** arranged parallel to each other. The multiple second rectangular structures **62** are substantially perpendicular to the multiple first rectangular structures **60**. The first rectangular structure **60** may include two U-shaped bars **64** disposed opposite and connected to each other. The second rectangular structure **62** may include two U-shaped bars **66** disposed opposite and connected to each other.

(37) FIG. **13** is an enlarged schematic view of a column joint **12**, in accordance with the embodiment of the present disclosure. As shown in FIG. **13**, the multiple first sets of beam bars **3** and the multiple second beam bars **4** surround the column joint **12** in the accommodating space S extending from the top of the column **10** (as shown in FIG. **1**). FIG. **14** shows multiple first rectangular structures **60** along the second direction D**2** and multiple second rectangular structures **62** along the first direction D**1** from the outer sides of the multiple first rebar cages **30** and the outer sides of the multiple second rebar cages **40** to surround the lateral sides of the multiple joints **12**.

(38) FIG. **15** is a schematic view showing a waffle mold **7**, in accordance with the embodiment of the present disclosure. As shown in FIG. **15**, the waffle mold **7** includes a hollow body **70** and multiple hollow cylinders **72** disposed on the top surface **74** of the hollow body **70**. In this embodiment, four hollow cylinders **72** are evenly arranged on the hollow body **70** in a **2***x***2** pattern. The area of the bottom surface **76** of the hollow body **70** is larger than the area of the top surface **74**, and the cross-sectional area of the hollow body **70** gradually decreases from its bottom to its top, forming a wedge-shaped quadrilateral.

(39) FIG. **16** is a still further schematic view showing the embodiment of the present disclosure. FIG. **16** shows multiple waffle molds **7** provided in multiple accommodating spaces S defined by the multiple first sets of beam bars **3** and the multiple second sets of beam bars **4**. In this embodiment, no waffle mold **7** is provided at the column joint **12**.

(40) FIG. **17** is a still further schematic view showing the embodiment of the present disclosure. FIG. **17** shows multiple reinforcing bars **8** arranged between the multiple hollow cylinders **72** of a waffle mold **7**, and extending between the hollow cylinders **72** of the adjacent waffle molds **7**. In addition, the multiple reinforcing bars **8** span over the multiple first sets of beam bars **3** or the multiple second sets of beam bars **4**.

(41) FIG. **18** is a schematically perspective view showing the embodiment of the present disclosure. FIG. **18** shows concrete **9** poured into the molds (not shown) surrounding the plane and solidified so that the concrete **9** covers the aforementioned multiple first sets of beam bars **3**, the multiple second sets of beam bars **4** and the multiple waffle molds **7**. After the concrete **9** has solidified, the bottom mold **1** and the multiple waffle molds **7** are removed to form the waffle slab **100**. As shown in FIG. **18**, the waffle slab **100** includes a top plate portion **102**, multiple supporting ribs **104** and supporting blocks **106**. The supporting ribs **104** and the supporting blocks **106** are located on the bottom surface of the top plate portion **102**. The positions of the supporting blocks **106** correspond to the positions of the columns **10** below. The multiple supporting ribs **104** extend along the first direction D**1** and the second direction D**2**, and form a recess R therebetween. The top plate portion **100** has multiple through holes **108**. In this embodiment, the location of each of the recesses R corresponds to that of a row of waffle molds **7** in the first direction D**1** or the second direction D**2** before the waffle molds **7** are removed.

(42) FIG. **19** is a schematic bottom view of the structure shown in FIG. **18**. FIG. **20** is a partial cross-sectional view of FIG. **18** along line A-A. As shown in FIG. **19** and FIG. **20**, the opening **24** of the elongated trough-shaped body **20** of the suspension rail **2** is configured to be exposed from the bottom surface of the supporting bar **104** for connecting other devices.

(43) The term “a” or “an” herein is used to describe elements and components of the present disclosure. This term is only for the convenience of description and to give the basic concept of this disclosure. Such a statement should be read to include one or at least one, and the singular also includes the plural unless it is clearly stated otherwise. When used together with the word “comprising” in the claims, the word “a” may mean one or more than one. In addition, the term “or” herein means “and/or.”
(44) Unless otherwise specified, terms such as “above,” “below,” “upward,” “left,” “right,” “downward,” “body,” “base,” “vertical,” “horizontal,” “side,” “higher,” “lower,” “upper,” “above” and “below” are spatial descriptions intended to indicate the directions shown in the drawings. It should be understood that the spatial descriptions used herein are for illustrative purposes only, and that actual implementations of the structures described herein may be spatially configured in any relative orientation, such constraints not altering the advantages of the disclosed embodiments. For example, in descriptions of some embodiments, providing an element “on” another element may encompass situations where the former element is directly on (e.g., in physical contact with) the latter element as well as a situation where a plurality of intervening elements are disposed between the former element and the latter element.
(45) As used herein, the terms “approximately,” “substantially,” “substantial” and “about” are used to describe and allow for minor variations. When used in conjunction with an event or circumstance, these terms can mean both instances in which the event or circumstance expressly occurred as well as instances in which the event or circumstance occurred in close proximity.
(46) The present disclosure is not limited to the specific structure or arrangement disclosed herein, and those with ordinary knowledge in the art of the present disclosure should understand that, under the spirit of the present disclosure, these structures and arrangements disclosed herein must be within certain limits, and that the extent may be altered or substituted. It should also be understood that the terminology used herein and the terms describing directions or relative positions are only used to describe specific embodiments and facilitate explanation and understanding, and are not intended to limit the scope of the present disclosure.

Claims

1. A method of manufacturing a waffle slab on a plurality of columns at intervals, the method comprising: disposing a bottom mold separately on the plurality of columns and exposing joint heads of the plurality of columns from a top surface of the bottom mold; disposing a plurality of first sets of beam bars separately on the bottom mold so that the plurality of first sets of beam bars are parallel to each other along a first direction, wherein the plurality of first sets of beam bars each comprises: a plurality of first rebar cages disposed along the first direction at intervals, and a plurality of first upper rebars and a plurality of first lower rebars that both penetrate through an inside of the plurality of first rebar cages and are respectively affixed to upper portions and lower portions of the plurality of first rebar cages; disposing a plurality of second sets of beam bars separately on the bottom mold so that the plurality of second sets of beam bars are parallel to each other along a second direction, the second direction being different from the first direction, wherein the plurality of second sets of beam bars each comprises: a plurality of second rebar cages separately disposed along the second direction; and a plurality of second upper rebars penetrating through an inside of the plurality of second rebar cages and affixed to upper portions of plurality of the second rebar cages, wherein the plurality of second upper rebars are attached to parts of the plurality of first upper rebars that are located between two of the plurality of first rebar cages; and providing and penetrating a plurality of second lower rebars through an inside of the plurality of second rebar cages and affixing the plurality of second lower rebars to the lower portions of the plurality of the second rebar cages; wherein the plurality of second sets of beam bars are configured to intersect with the plurality of first sets of beam bars without interference, so that the plurality of

first sets of beam bars and the plurality of second sets of beam bars together form a plurality of accommodation spaces therebetween.
2. The method of claim 1, further comprising: providing a plurality of waffle molds into the plurality of accommodation spaces; pouring concrete to cover the plurality of first sets of beam bars, the plurality of second sets of beam bars and the plurality of accommodation spaces; and removing the bottom mold and the plurality of waffle molds after the concrete is solidified.
3. The method of claim 2, before the step of providing the plurality of first sets of beam bars and the plurality of second sets of beam bars, further comprising: providing a plurality of rails on the bottom mold along the second direction, wherein each of the plurality of rails includes an elongated trough-shaped body and a plurality of supports on the elongated trough-shaped body, wherein the elongated trough-shaped body has an opening therein and the plurality of supports are located on the opposite side of the opening; wherein the plurality of second sets of beam bars are disposed on the plurality of supports of the rails, and the opening of the elongated trough-shaped body is configured to face the bottom mold.
4. The method of claim 2, wherein the step of providing the plurality of first sets of beam bars further comprises: hoisting the plurality of first sets of beam bars at intervals simultaneously with a lifter, configured so that the heights of the plurality of first sets of beam bars are different; and disposing the plurality of first sets of beam bars on the bottom mold in order.
5. The method of claim 2, wherein the step of providing the plurality of second sets of beam bars further comprises: providing a plurality of couplers to connect segments of the plurality of second upper rebars in series in each of the plurality of second rebar cages.
6. A structure for manufacturing a waffle slab, comprising: a bottom mold disposed on a plurality of columns disposed at intervals, wherein a joint head of each of the plurality of columns is exposed from a top surface of the bottom mold; a plurality of first sets of beam bars disposed separately on the bottom mold that are parallel to each other along a first direction, wherein the plurality of first sets of beam bars each comprises: a plurality of first rebar cages disposed along the first direction at intervals, and a plurality of first upper rebars and a plurality of first lower rebars that both penetrate through an inside of the plurality of first rebar cages and are respectively affixed to upper portions and lower portions of the plurality of first rebar cages; a plurality of second sets of beam bars disposed separately on the bottom mold that are parallel to each other along a second direction, the second direction being different from the first direction, wherein the plurality of second sets of beam bars each comprises: a plurality of second rebar cages separately disposed along the second direction; and a plurality of second upper rebars and a plurality of second lower rebars that both penetrate through an inside of the plurality of second rebar cages and are respectively affixed to upper portions and lower portions of the plurality of second rebar cages, wherein the plurality of second upper rebars are attached to parts of the plurality of first upper rebars that are located between two of the plurality of first rebar cages; wherein the plurality of second sets of beam bars are configured to intersect with the plurality of first sets of beam bars without interference, so that the plurality of first sets of beam bars and the plurality of second sets of beam bars together form a plurality of accommodation spaces therebetween.
7. The structure of claim 6, further comprising a plurality of waffle molds disposed in the plurality of accommodation spaces.
8. The structure of claim 6, further comprising: a plurality of rails on the bottom mold along the second direction, wherein each of the plurality of rails includes an elongated trough-shaped body and a plurality of supports on the elongated trough-shaped body, wherein the elongated trough-shaped body has an opening and the plurality of supports are located on the opposite side of the opening; wherein the plurality of second beam sets are disposed on the plurality of supports of the rails, and the opening of the elongated trough-shaped body is configured to face the bottom mold.
9. The structure of claim 6, further comprising: a plurality of waist bars disposed between and being generally parallel with the plurality of second upper rebars and the plurality of second lower

rebars, and; a plurality of tie bars configured to secure the positions of the plurality of waist bars.
10. The structure of claim 6, further comprising: a plurality of column tie bars including a plurality of first rectangular structures disposed parallel to each other and a plurality of second rectangular structures substantially vertically disposed and fixed to the plurality of first rectangular structures; wherein the plurality of first rectangular structures and the plurality of second rectangular structures are configured to surround sides of the joint heads.
11. The structure of claim 10, wherein the plurality of first rectangular structures and the plurality of second rectangular structures are configured to be secured to the portions of the plurality of first sets of beam bars and the plurality of second sets of beam bars that surround the joint heads.
12. The structure of claim 10, wherein each of the plurality of first rectangular structures and the plurality of second rectangular structures are formed of two U-shaped bars connected to each other.
13. The structure of claim 6, wherein the second direction is generally perpendicular to the first direction.
14. The structure of claim 7, wherein the plurality of waffle molds includes a hollow body and a plurality of hollow cylinders disposed on the hollow body, and the structure further comprises a plurality of reinforcing bars disposed on the plurality of waffle molds between the hollow cylinders.